I_Phi Landscape Analysis — Hubbard L=6 Ne=6 t=1.0 U=4.0 mode=uniform

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System
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  L=6  Ne=6  t=1.0  U=4.0  mode=uniform
  n = N = 6  (pure-state RDM, Tr(rho^2)=1)

  E0  = -3.6687061789
  E1  = -2.8983814740
  Gap = 0.77032470

GME Results
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  C_GME (exact eq.1)   = 0.95823220
  C_LB  (basis states) = 0.45569925
  Gap   = C_GME - C_LB = 0.50253294

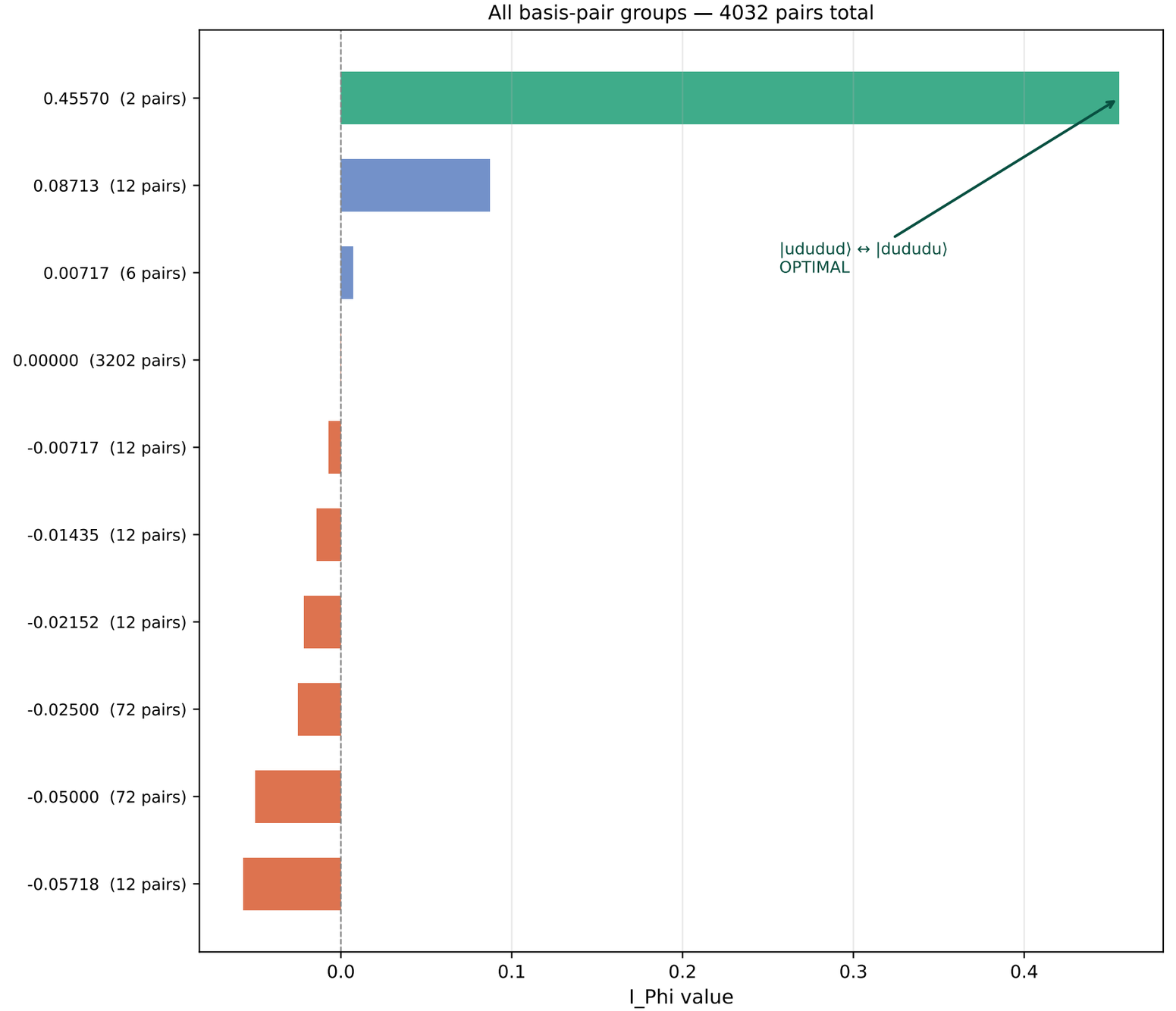
Optimal |Phi>
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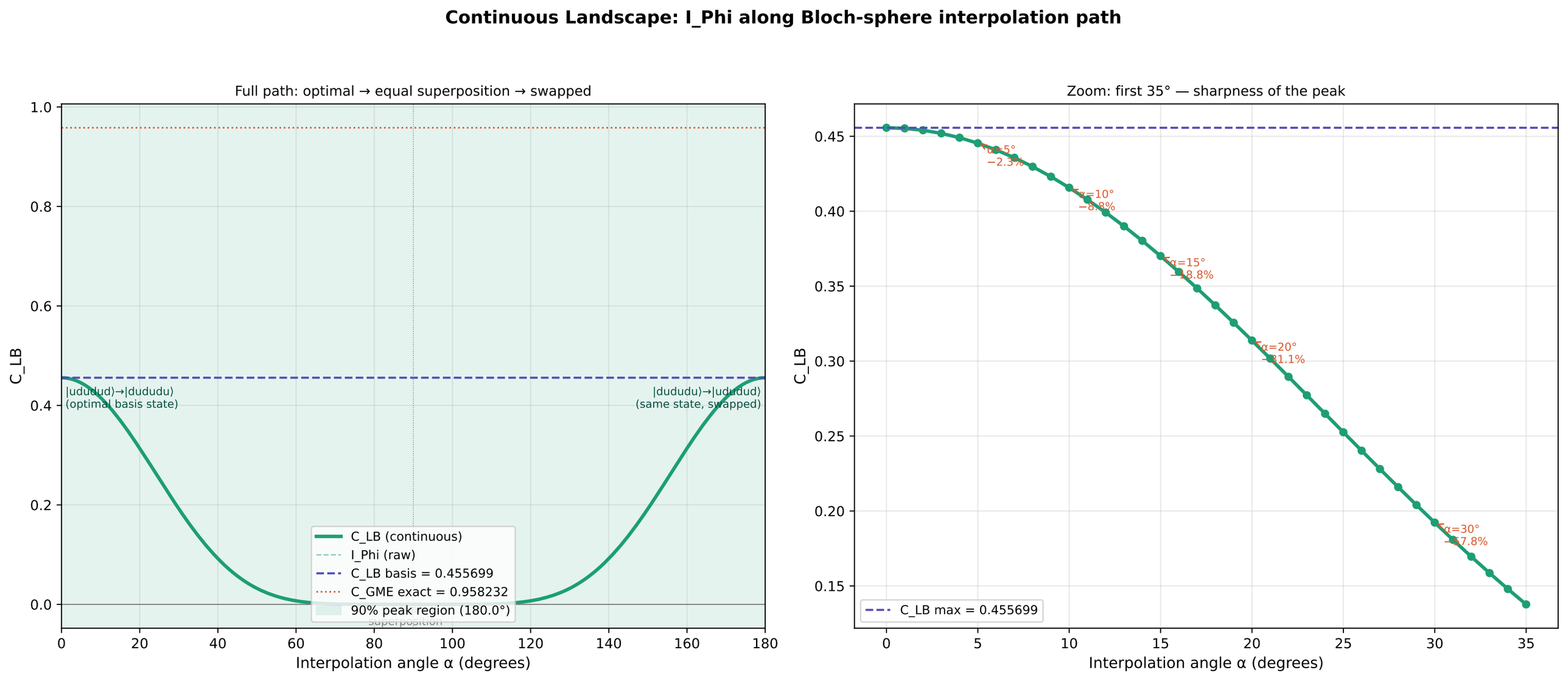
  z1 = |ududud⟩
  z2 = |dududu⟩

  Pairs at maximum     : 2
  Physically unique    : 1
  (z1→z2 and z2→z1 are
   the same |Phi⟩, just
   labels swapped)

Continuous Landscape
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  Peak width at 90%    : 180.0°
  Peak character       : BROAD
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Continuous Landscape: I_Phi along Bloch-sphere interpolation path
Full path: optimal → equal superposition → swapped
C_LB
Interpolation angle α (degrees)
|udududud⟩→|dududu⟩
(optimal basis state)
|dududu⟩→|udududud⟩
(same state, swapped)
Superposition
C_LB (continuous)
I_Phi (raw)
C_LB basis = 0.455699
C_GME exact = 0.958232
90% peak region (180.0°)
Zoom: first 35° — sharpness of the peak
C_LB
Interpolation angle α (degrees)
α=5°
−2.3%
α=10°
−8.8%
α=15°
−18.8%
α=20°
−31.1%
α=30°
−57.8%
C_LB max = 0.455699

Summary: Answer to Supervisor's Question

Supervisor's question:
  'Is the maximum obtained only for the basis state, or do you also reach
   it for other states Phi? Are there single optimal points in the parameter
   space, or multiple optimal points?'

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1. DISCRETE (basis-state) SEARCH [4032 pairs checked]

   2 pairs achieve the maximum I_phi = 0.45569925:
     |ududud⟩ → |dududu⟩
     |dududu⟩ → |ududud⟩

   These are physically IDENTICAL — z1 and z2 labels swapped.
   -> ONE unique optimal |Phi⟩ in basis-state space.

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2. CONTINUOUS LANDSCAPE (real states, Bloch-sphere interpolation)

   The peak is SHARP and ISOLATED at the basis state:

   alpha =   5°   C_LB = 0.445393   (+2.3% from max)
   alpha = 10°   C_LB = 0.415707   (+8.8% from max)
   alpha = 15°   C_LB = 0.370119   (+18.8% from max)
   alpha = 20°   C_LB = 0.313757   (+31.1% from max)
   alpha = 30°   C_LB = 0.192248   (+57.8% from max)
   alpha = 45°   C_LB = 0.056962   (+87.5% from max)
   alpha = 90°   C_LB = 0.000000   (+100.0% from max)

   Peak width at 90% of maximum: 180.0°
   -> BROAD peak

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CONCLUSION:

   SINGLE isolated optimal point in parameter space.
   The basis-state |ududud⟩ ↔ |dududu⟩ is the unique global maximum.
   Moving even a few degrees away on the Bloch sphere reduces I_Phi.
   No other family of states achieves the same value.

   This confirms: the Hubbard ground state has Neel-type entanglement
   concentrated in the coherence between the two antiferromagnetic
   configurations. The basis-state approach is already exactly optimal.